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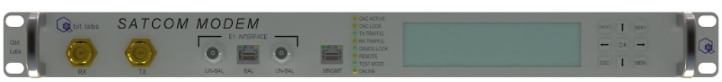
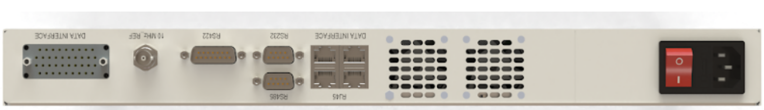
Qbit Advanced Satellite Modem (QASM-01)

The Satellite Modem is a compact communication system designed for efficient and reliable data transmission over satellite links. It integrates advanced modulation, forward error correction (FEC), and adaptive coding technologies to maintain link performance under varying conditions.
The modem supports flexible data rates, multiple interfaces, and dual-band operation, along with features such as carrier-in-carrier, bandwidth optimization, packet processing, QoS control, and secure data encryption. Its scalable architecture and remote management capabilities make it suitable for applications like IP trunking, mobile backhaul, and enterprise satellite communications.

Introduction

The Satellite Modem is a high-performance communication system designed to provide reliable and efficient data transmission over satellite links for telecom, enterprise, and mission-critical applications. It enables secure, real-time connectivity by integrating advanced RF interfacing, flexible modulation schemes, and powerful forward error correction techniques to maintain stable link performance under varying channel and environmental conditions.
The modem incorporates sophisticated baseband processing, adaptive coding and modulation, and bandwidth optimization technologies to maximize spectral efficiency, improve availability, and support a wide range of data rates and interfaces. With integrated packet processing, quality-of-service control, encryption, and remote management capabilities, it offers seamless integration into modern IP-based satellite networks.
Built with a scalable and modular architecture, the system supports dual-band operation, carrier-in-carrier optimization, and comprehensive monitoring and control features, making it well suited for applications such as mobile backhaul, IP trunking, enterprise connectivity, and secure satellite communications.
Advance Carrier-in-Carrier feature allows sending and receiving of satellite data across same carrier frequency for either power limited or bandwidth limited scenarios





Architecture

The Satellite Modem architecture is designed around an integrated RF, baseband, and networking processing framework to ensure efficient and reliable satellite communication. It consists of a dual-band RF interface supporting standard IF and L-Band operation, enabling flexible transmit and receive configurations. The RF section is followed by high-precision modulation and demodulation blocks that support multiple modulation schemes and adaptive filtering for stable signal acquisition and tracking.
At the core of the system, FPGA/ASIC-based baseband processing performs functions such as adaptive coding and modulation (ACM), forward error correction (FEC), synchronization, equalization, and carrier recovery to ensure accurate data transmission with high spectral efficiency. Advanced bandwidth optimization features, including carrier-in-carrier technology and compression mechanisms, further enhance link performance.
The architecture also integrates a packet processing and networking subsystem with an embedded Ethernet switch, QoS management, header and payload compression, and encryption support. A dedicated management and control module provides remote monitoring, configuration, synchronization, and diagnostics, enabling seamless integration into satellite communication networks.

Why Choose Us?

Rapid integration, faster time-to-market

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Direct access to experts

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Features

- Supports wide data rates and symbol rates for flexible satellite link operation.
- Multiple modulation schemes including BPSK, QPSK, OQPSK, 8PSK/8-QAM, and 16-QAM.
- Advanced Forward Error Correction (FEC) such as Viterbi, RS Decoder, LDPC, Turbo Product Codes.
- Adaptive coding & modulation (ACM) and constant coding and modulation (CCM).
- Advanced Equalization scheme with MMSE equalization.
- Carrier-in-Carrier bandwidth optimization to improve spectral efficiency and reduce operating costs.
- Dual-band capability supporting standard IF and extended L-Band operation in a single unit.
- Integrated packet processor with header and payload compression for efficient IP transmission.
- Built-in multi-port managed Ethernet switch with VLAN and Quality of Service (QoS) support.
- AES-based data encryption for secure satellite communications.
- Automatic uplink power control and adaptive equalization for reliable link performance.
- Multiple data interfaces including Ethernet, serial, and G.703 telecom interfaces.
- Comprehensive monitoring, control, and remote management via SNMP, HTTP, and serial interfaces.
- Support for synchronization features such as precision time protocol and reference clock inputs.
- Modular and scalable architecture enabling redundancy options and field-upgradeable features.

Specifications

Parameter	Value
Data Rate	• 18 kbps to 8 Mbps in 1 bps steps
Operating Frequency	• IF: 50 – 180 MHz • L-Band: 950 – 2250 MHz • Independent TX and RX operation with 100Hz resolution
Major Operating Modes	• Closed network • LDPC / TPC Codec • Carrier-in-Carrier
FEC & Modulation	
Parameter	**Value**
None	• Uncoded BPSK/QPSK/OQPSK
Viterbi	• Rate 1/2 BPSK/QPSK/OQPSK • Rate 2/3, 3/4 QPSK/OQPSK • Rate 7/8 QPSK/OQPSK
Integrated LDPC & TPC Codec	**LDPC Code Rates:** • Rate 1/2 BPSK/QPSK/OQPSK; • Rate 2/3 QPSK/OQPSK/8PSK/8-QAM; • Rate 3/4 QPSK/OQPSK/8PSK/8-QAM/16-QAM **TPC Code Rates:** • Rate 5/16 BPSK; • Rate 21/44 BPSK/QPSK/OQPSK; • Rate 3/4 QPSK/OQPSK/8PSK/8-QAM/16-QAM; • Rate 7/8 QPSK/OQPSK/8PSK/8-QAM/16-QAM; • Rate 0.95 QPSK/OQPSK/8PSK/8-QAM

Viterbi with Reed Solomon	• Rate 3/4 and 7/8 16QAM
Reed Solomon	• RS(255,239), RS(255,223)
Data Interfaces	• 4-port 10/100 Base-T Managed Ethernet Switch with 4 RJ-45 Ports. • EIA-422, V.35 DCE, G.703 E1 2.048Mbps (Unbalanced 75 Ω or balanced 120 Ω)
Modulator	
Parameter	**Value**
Transmit Filtering	• With Roll off 5%, 10%, 15%, 20%, 25%, 35%
Harmonics and Spurious	• −60 dBc/4 kHz
Transmit On/Off Ratio	• −65 dBc
Output Power	• IF: 0 to −25 dBm, 0.1 dB steps • 950–2250 MHz: +5 to -30dBm, 0.1 dB steps
Power Accuracy	• ± 1.0 dB, 0 to 50 deg C
Output Impedance & Return Loss	• 50–180 MHz: 50-ohm, 18 dB return loss • 950–2250 MHz: 50-ohm, 15 dB return loss
Clock Options	• Internal: 50 ppb • External Clock:locking over a ±100 ppm range (TT) • Loop timing (Rx satellite clock) – Support asymmetric operation
BUC Reference (10 MHz)	• Via TX IF Centre conductor • 10.0 MHz ± 0.01 ppm (with internal reference), selectable on/off, • Power output of 0.0 dBm ± 3 dB
Demodulator	
Parameter	**Value**
Input Power Range, Desired Carrier	• 50–180 MHz: −135 + 10 log (symbol rate) to −68 + 10 log (symbol rate) dBm • 950–2250 MHz: −145 + 10 log (symbol rate) to −68 + 10 log (symbol rate) dBm
Maximum Input Power	• +10 dBm
Output Impedance & Return Loss	• 50–180 MHz: 50 ohm, 18 dB return loss • 950–2250 MHz: 50 ohm, 15 dB return loss

Receive Clock	• Rx Satellite, Tx Terrestrial, External Reference
Clock Tracking	• ±100 ppm
LNB Reference (10 MHz)	• Via RX IF Centre conductor • 10.0 MHz ± 0.01 ppm (with internal reference), selectable on/off, 0.0 dBm ± 3 dB
LNB Voltage	• Selectable on/off 13/18/24 VDC
Monitor Functions	• Eb/No estimate, Correct BER, Frequency offset, Buffer fill state, Receive signal strength (RSSI)
Bandwidth Save Option	
Parameter	**Value**
Carrier-in-Carrier / Paired Carrier Board	• Data Rate 18 kbps to 8 Mbps full
Environmental and Physical Specifications	
Parameter	**Value**
Temperature	• Operating: 0 to 50°C, 95% Humidity • Storage: 0 to 70°C, 95% Humidity
Power Supply	• 100–240 VAC, 50 to 60 Hz
Physical Dimensions	• 1RU, 19-inch rack mountable • Mounting Kit is provided

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